

Harshit Kumar

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Skills

- **AI & ML:** Generative AI | NLP | TensorFlow Lite | scikit-learn | Gemini 2.5 API | Groq API | Whisper API | FastAPI | CNN
- **Programming & Fundamentals:** Python | Java | C | SQL | OOPS | DSA | DBMS
- **Web & Backend Development:** Node.js | Express | Relational Database Management Systems (RDBMS) | SQL Schema Design
- **Voice & Cloud:** Google Cloud | AWS (EC2, S3, Lambda, IAM, API Gateway)| Vertex AI | Streamlit | API Integration
- **Cybersecurity:** Cryptography | Network Security | Penetration Testing
- **Core:** Leadership | AI Product Development | Problem Solving | Prototyping

Education

Bachelor of Technology - Computer Science Engineering

Aug 2024 – Present

Vellore Institute of Technology, A.P C.G.P.A- 8.23/10.0

Intermediate – PCM

April 2022-May-2023

St. Anthony Public School ,U.P Percentage – 77.6%

Secondary School- CBSE

St. Anthony Public School , UP Percentage – 89.8%

March 2020- May-2021

Projects

Smart Classroom & Timetable Scheduler

- **Tech Stack:** Python / Node.js, SQL, Constraint Satisfaction Algorithms
- **Key Bullet Points:**
 - Developed an intelligent scheduling platform using constraint satisfaction logic to generate conflict-free academic timetables across multiple departments.
 - Optimized faculty-room allocation workflows, improving spatial and scheduling efficiency in higher education environments.

Wafer AI Map Detection — Explainable Semiconductor Defect Classifier

- **Tech Stack :** Python, PyTorch/TensorFlow, Grad-CAM, Gradio, OpenCV
- **Engineered End-to-End Visual Inspection Pipeline:** Built a Deep Convolutional Neural Network (CNN) to automate spatial defect classification (e.g., Scratch, Edge-Ring, Center, Local) in semiconductor wafer maps.
- **Implemented Explainable AI (XAI):** Integrated Gradient-weighted Class Activation Mapping (GradCAM) to generate real-time activation heatmaps, providing transparent defect localization for rootcause yield analysis.
- **Built Interactive Dashboard & Reporting:** Developed a live web UI using Gradio featuring dynamic reliability scoring and automated PDF inspection report generation for quality assurance reviews.

Achievements

Google Gen AI Academy (APAC Edition) — Tracks 1 & 2

- **Core Focus:** Advanced Generative AI development, agentic workflows, and Google Cloud Platform (GCP) infrastructure.
- **Key Competencies:** Hands-on experience building multi-agent AI systems, prompt engineering, containerizing
- backend services using Docker, and deploying serverless AI applications to GCP Cloud Run.

Winner — Null Matrix CTF (Cybersecurity Hackathon):

- Secured top ranking by solving complex challenges spanning Web Exploitation, Cryptography, Reverse Engineering, and Network Forensics. Applied hands-on security analysis and vulnerability assessment techniques under timed constraints.